MILITARY SPECIFICATION

INTERVALOMETER ADAPTER FOR M56 AIRCRAFT SUBSYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection and test of the intervalometer adapter used in the M56 Aircraft Sub-system.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition

FSC: 1345

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment for Inert Ammunition Components.
MIL-STD-1235 - Single and Multilevel Continuous Sampling.

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT DRAWING

9280150 - Adapter, Intervalometer

INSPECTION EQUIPMENT DRAWING

9278638 - M56 Intervalometer Adapter Test Set.

(Copies of specifications, standards, drawings and publications listed above or in the item detail specification required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable specifications and drawings.

3.2 Part. - The part shall comply with all requirements specified on drawing (dwg.) 9280150 and with all requirements specified in applicable specifications.

3.3 Electrical Continuity. - The intervalometer shall have a maximum resistance of five hundredths (0.05) of an ohm between corresponding pins on connectors J1 and 32 where a direct connection is made. In addition, this continuity requirement shall exist between pin DD of J1 and pins AA and EE of J2. Continuity as defined shall not exist between pin u of J1 or pin y of J2 and either pin DD of J1 or pin DD of 32.

3.4 Diode forward voltage drop. - The diode forward voltage drop shall be obtained at either pin u of J1 or y of J2 when pin DD of J1 or J2 is driven with a current of one (1) ampere plus or minus ten percent (± 10%).

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3.5 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, excessive epoxy or sealer build-ups, loose wire strands, exposed leads on internal pins or solder joints and other foreign matter. The cleaning agent or method used shall not be Injurious to any part, nor shall the part be contaminated by the cleaning agents. All markings and stampings shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests. - The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. Results shall be submitted within thirty (30) days at the completion of each lot tested (see 6.4).

3.7 First Article Testing. - This specification makes provisions for First Article Testing. Submission of the First Article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following Information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A-statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by-the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of twenty-five (25) sets of each sub-assembly and twenty five (25) complete assemblies. Which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this Inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or spec-

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ification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and sub-assemblies will be subjected by the Government to any or all of the examinations and tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any component or subassembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or subassembly to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see

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4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 - percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Connector (J1) Modified (See dwg. 9281037 covering a detail of 9280150).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40%		
101.	Modified thread diameter .	Gage	01001
102.	Length of modified thread diameter	Gage	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	01003

4.3.2.2 "W" Cable Seal, Modified (see dwg. 9281040 covering a detail of 9280150).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40%		
101.	Modified length	Gage	02001

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	02002

4.3.2.3 Adapter, Intervalometer, J2 Connected only (see dwg. 9280150).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Sleeving covering all exposed wires and connectors (see dwg. note 4) ...	Visual	03001
102.	Locking compound on "W" cable seal	Visual	03002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	03003

4.3.2.4 Adapter, Intervalometer, J1 and J2 Wired (see dwg. 9280150)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Sleeving covering all exposed wires and connections (see dwg. note 4) ...	Visual	04001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	04002

4.3.2.5 Adapter, Intervalometer, Complete (see dwg. 9280150).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:		AQL 0.40 percent	
101.	Epoxy on J-1 to "W" cable...	Visual	05002
Minor:		AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.5)	Visual	05002

4.3.3 Testing

4.3.3.1 J2 Push Test (see dwg. 92801050), Major Defect, Defect No. 03004. - Each connector assembly prior to wiring shall be tested for the specified requirement in accordance with 4.4.1. Any sub-assembly classed defective shall be removed from the lot.

4.3.3.2 J1 Torquing (see dwg. 9280150), Major Defect, Defect No. 05003. - Each assembly shall be tested for the specified requirement in accordance with 4.4.2. Any assembly classed defective shall be removed from the lot.

4.3.3.3 Continuity Test (see paragraph 3.3), Major Defect, Defect No. 05004. - Each connector shall be tested for the specified requirements in accordance with 4.4.3. Any assembly classed defective shall be removed from the lot.

4.3.3.4 Diode Forward Voltage Drop. - (see paragraph 3.4), Major Defect, Defect No. 05005. Each assembly shall be tested for the specified requirements in accordance with 4.4.4. Any assembly classed defective shall be removed from the lot.

4.3.4 Inspection Equipment. - Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approval.

4.4 Test Methods and Procedures

4.4.1 J2 Push Test. - The connector assembly shall be mounted in an approved fixture and pushed from the outside with the specified force requirement. The connector shall be monitored for displacement during the entire time the force is applied to J2 to ascertain any movement. Any assembly failing

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to comply with all of the specified requirements shall be classed defective. Testing is non-destructive and only accepted assemblies shall be returned to the lot.

4.4.2 J1 Torquing. - The assembly shall be tested for the specified requirements using an approved fixture and torque wrench with an approved adapter socket. The assembly shall be monitored for movement during the time the torque is applied to ascertain if the bond is secured. Any assembly failing to comply with all of the requirements shall be classed defective. Testing is non-destructive and only accepted assemblies shall be placed into the lot.

4.4.3 Continuity Test. - The continuity test shall be performed on the intervalometer adapter using the test set shown on drawing 9278638. The test method is as follows:

1. Connect plug and connector together.
2. Zero out ohmmeter on J1 B and J2 B.
3. Place device under test (DUT) into test set between J1 A and J2 A.
4. Measure resistance between all points on J1 B and J2 B.

Equipment Necessary:

Digital ohmmeter

J2A Bendix Corp. Connector No. PT06W-22-55S
J1A Bendix Corp. Connection No. PT01A-22-55P

Alternate Method:

The resistance may be measured using a three deck stepper switch instead of manually using the ohmmeter. Any device failing to comply with all of the requirements of 3.3 shall be classed defective and removed from the lot. Testing is non-destructive and only accepted units shall be returned to the lot.

4.4.4 Diode Forward Voltage Drop. - The test shall be performed on the intervalometer adapter using the test set shown in drawing 9278638 by:

1. Driving the diode with one (1) ampere by measuring the current through meter, M1.

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2. Measure voltage drop across diode by pressing switch S1 and reading voltage on meter M2.

Equipment Needed:

1. Power supply
2. 5 - Ampere range d.c. ammeter
3. 5 - volt range d.c. voltmeter
4. Momentary contact, normally open switch.

Any device failing to comply with all of the applicable requirements of 3.4 shall be classed defective and removed from the lot. Testing is non-destructive and only accepted units shall be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C. - The intervalometer adapter shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing

5.2.1 Level C. - The intervalometer adapter shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. - Marking of the carton shall comply with MIL-sTD-1169.

5.4 Shipping. - When intervalometer adapters of more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly marked and indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2- Contractor designs. -Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with

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commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission results of contractor conducted examinations and tests (see 3.6). - The contractor shall send his results to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-M
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-033.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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